

```
1 import java.util.*;;
2 class Queue
3 {
4     int que[ ];
5     int size,front,rear;
6
7     Queue(int n)
8     {
9         size=n;
10        que = new int[size];
11        front=-1;
12        rear=-1;
13    }
14
15    void insert(int x)
16    {
17        if(rear == size-1)
18            System.out.println("Overflow");
19        else if(front==-1 && rear==-1)
20        {
21            front++;
22            rear++;
23            que[rear]=x;
24        }
25        else
26        {
27            rear++;
28            que[rear]=x;
29        }
30    }
31
32    int delete()
33    {
34        if(front==-1 && rear==-1)
35        {
36            System.out.println("Underflow");
37            return -9999;
38        }
39        else if(front==rear)
40        {
41            int temp=que[front];
42            front=-1;
43            rear=-1;
44            return temp;
45        }
46        else
47        {
48            int temp=que[front];
49            front++;
50            return temp;
```

```
51     }
52 }
53
54 public void display()
55 {
56     if(front==-1 && rear==-1)
57         System.out.println("Empty queue");
58
59     else
60     {
61         System.out.println("The queue is...");
62         for(int i=front;i<=rear;i++)
63             System.out.print(que[i]+" | ");
64     }
65     System.out.println("\n");
66 }
67
68 public static void main()
69 {
70     Scanner sc=new Scanner(System.in);
71     System.out.println("Enter the queue size");
72     int s=sc.nextInt();
73     int t;
74     Queue obj=new Queue(s);
75     int k,ch,element;
76
77     do
78     {
79         System.out.println("1. Add");
80         System.out.println("2. Remove");
81         System.out.println("3. Display");
82         System.out.println("4. Quit");
83         System.out.println("Enter choice ?");
84         ch=sc.nextInt();
85
86         switch(ch)
87         {
88             case 1: System.out.println("Enter the new element to be added at
89 rear");
90                 element=sc.nextInt();
91                 obj.insert(element);
92                 break;
93
94             case 2: t=obj.delete();
95                 if(t!=-9999)
96                     System.out.println("Deleted "+t);
97                 break;
98
99             case 3: obj.display();
100                 break;
```

```
101         case 4: System.exit(0);
102
103         default:
104             System.out.println("Wrong choice... Re enter your choice");
105     }
106
107     System.out.println("Press 1 to continue and any other number to
108 terminate");
109     k=sc.nextInt();
110     } while(k==1);
111 }
112 }
```